

Arithmetic nonsingularity of the odd receiving frame

positive integral ES witness, prime $p \equiv 1 \pmod{12}$

$N = S^6 F \in \mathbb{Z}$; test $N \neq 0$ before dividing by S

exact divisibility split

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Type I: $x, y \in \mathbb{Z}_p^\times$, $z = pZ$

$\not\equiv y$: E2, discriminant -511 ; $v_p(N) = 0$

$y = x + pw$: E3, square class 1241; $v_p(N) = 2$

The diagonal is calculated, not discarded.

Type II: $x = pX$, $y = pY$, $z = Z$

positivity forces $v_p(x) = v_p(y) = 1$

E4: $N/p^2 \equiv -5Z^6(4t^2 - 7t + 4)$

discriminant -15 ; hence $v_p(N) = 2$

$$\left(\frac{-511}{p}\right) = \left(\frac{1241}{p}\right) = \left(\frac{-15}{p}\right) = -1$$

$\Rightarrow N \neq 0 \Rightarrow \det O = 4\chi N/S^3 \neq 0$ for every witness at that prime

quadratic reciprocity gives 3456 reduced classes modulo 521220

The five-prime, 163-witness census checks the valuations; it is not the quantified proof.

Failure of one character condition is not a singularity certificate.